

Gallium

Ga (Ga I). Strongest classified transitions below 2 μm (NIST): 19 levels, 25 lines.

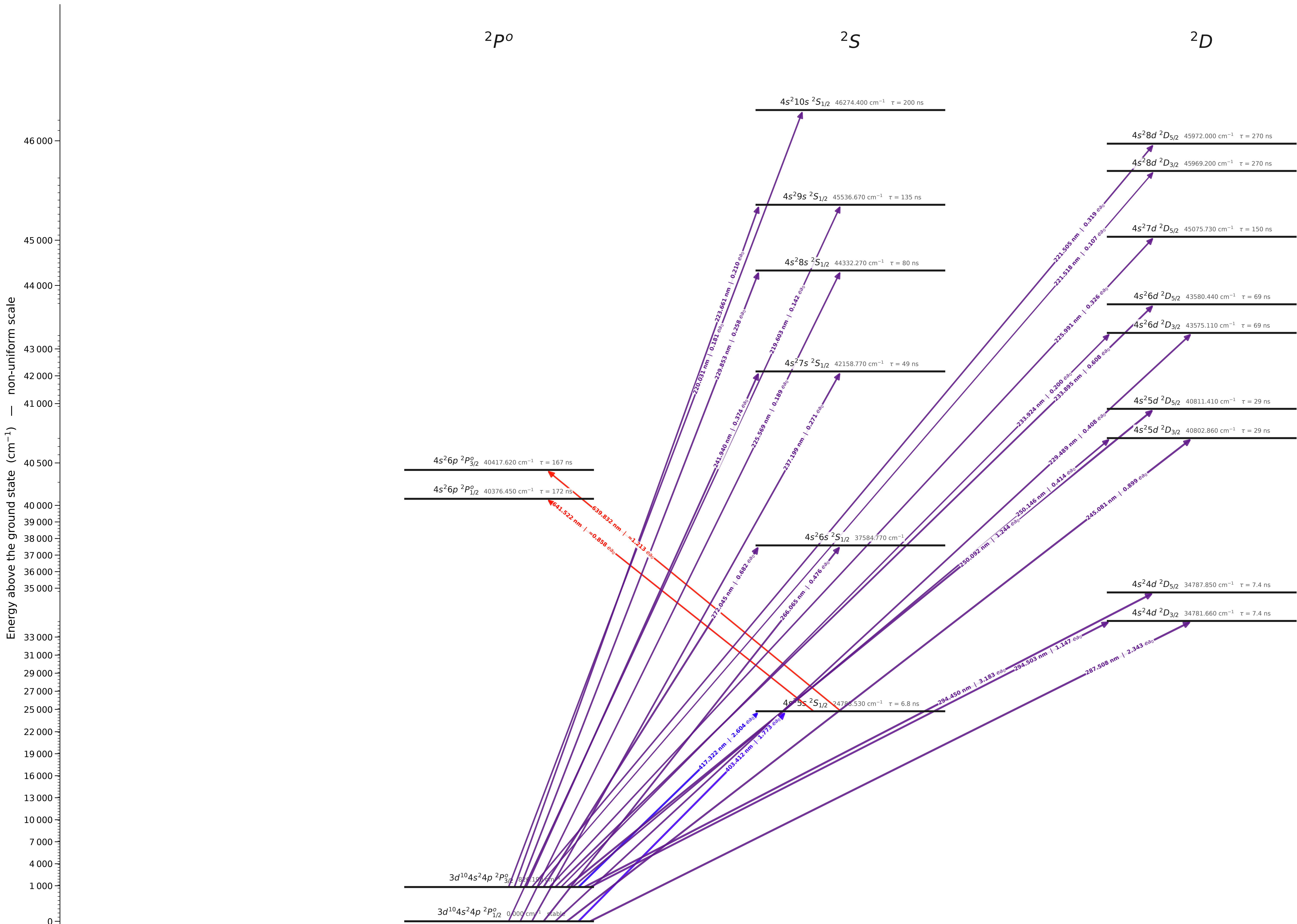
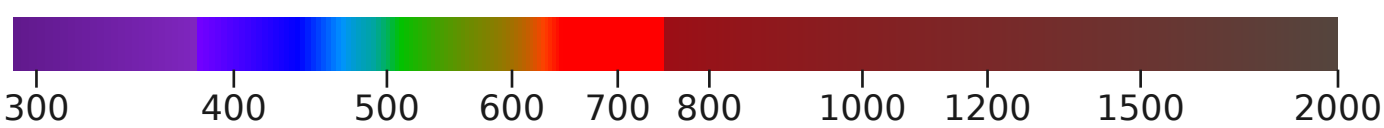
Ga⁺ ionisation limit 48387.63 cm⁻¹ = 5.99930 eV (206.664 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J \| e r \| J' \rangle|$ in ea_0

Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model or semi-empirical calculation.

Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J er J\rangle $ (ea0)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$3d^{10}4s^24p \ ^2P_{1/2}^o - 4s^24d \ ^2D_{3/2}$	287.5078	287.4235	1042.7279	2.343	1.170e+08	21.51	86.6	287.4 nm
$3d^{10}4s^24p \ ^2P_{3/2}^o - 4s^24d \ ^2D_{5/2}$	294.4497	294.3636	1018.1449	3.183	1.340e+08	21.51	99.2	294.45 nm laser-cooling
$3d^{10}4s^24p \ ^2P_{1/2}^o - 4s^25s \ ^2S_{1/2}$	403.4124	403.2984	743.1414	1.773	4.850e+07	23.41	33	403.3 nm resonance line
$3d^{10}4s^24p \ ^2P_{3/2}^o - 4s^25s \ ^2S_{1/2}$	417.3215	417.2039	718.3729	2.604	9.450e+07	23.41	64.3	417.2 nm line
$4s^25s \ ^2S_{1/2} - 4s^26p \ ^2P_{3/2}^o$	639.8325	639.6557	468.5483	≈ 1.213	2.845e+06	0.953	47.5	639.7 nm
$4s^25s \ ^2S_{1/2} - 4s^26p \ ^2P_{1/2}^o$	641.5224	641.3451	467.3141	≈ 0.858	2.822e+06	0.9253	48.5	641.3 nm

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

$\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

⁶⁹Ga hyperfine structure (*I* = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$3d^{10}4s^24p^2P_{1/2}^o$	1338.9938(5)	-	F=1: -1673.74 F=2: +1004.25
$3d^{10}4s^24p^2P_{3/2}^o$	191.5(9)	63.0(20)	F=0: -639.38 F=1: -510.88 F=2: -190.88 F=3: +446.62
$4s^25s^22S_{1/2}$	1069.5(15)	-	F=1: -1336.88 F=2: +802.12
$4s^24d^22D_{3/2}$	-36.3(22)	5(4)	F=0: +141.88 F=1: +100.97 F=2: +23.77 F=3: -80.53
$4s^24d^22D_{5/2}$	77.3(9)	5(4)	F=1: -402.12 F=2: -251.75 F=3: -22.24 F=4: +291.20
$4s^26p^22P_{1/2}^o$	53.3(7)	-	F=1: -66.62 F=2: +39.98
$4s^26p^22P_{3/2}^o$	22.40(20)	3.2(3)	F=0: -80.00 F=1: -60.80 F=2: -19.20 F=3: +51.20

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$3d^{10}4s^24p^2P_{3/2}^o - 4s^24d^2D_{3/2}$	287.508	71-69	123(20)
$3d^{10}4s^24p^2P_{3/2}^o - 4s^24d^2D_{5/2}$	294.450	71-69	115(7)
$3d^{10}4s^24p^2P_{3/2}^o - 4s^24d^2D_{3/2}$	294.503	71-69	114(8)
$3d^{10}4s^24p^2P_{1/2}^o - 4s^25s^2S_{1/2}$	403.412	69-71	33(4)
$3d^{10}4s^24p^2P_{3/2}^o - 4s^25s^2S_{1/2}$	417.322	69-71	40(4)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$